

Solved Example: Cyclomatic Complexity & Basis Paths

1 Notation (Exam-Safe)

- N = number of nodes in the control flow graph (CFG)
- E = number of edges in the CFG
- P = number of **connected components** in the CFG (usually $P = 1$)
- P_d = number of **predicate/decision nodes** (if, while, for, etc.)
- R = number of **regions** in the CFG (including the outside region)

Cyclomatic Complexity (McCabe):

$$V(G) = E - N + 2P, \quad V(G) = R, \quad V(G) = P_d + 1 \text{ (structured programs)}$$

2 Given Problem

2.1 Code (Typed from the diagram)

```
INTERFACE RETURNS average, total.input, total.valid;
INTERFACE ACCEPTS value, minimum, maximum;

TYPE value[1:100] IS SCALAR ARRAY;
TYPE average, total.input, total.valid,
    minimum, maximum, sum IS SCALAR;
TYPE i IS INTEGER;

i = 1;
total.input = total.valid = 0;
sum = 0;

DO WHILE value[i] <> -999 AND total.input < 100
    increment total.input by 1;
    IF value[i] >= minimum AND value[i] <= maximum
        THEN increment total.valid by 1;
            sum = sum + value[i]
        ELSE skip
    ENDIF
    increment i by 1;
ENDDO

IF total.valid > 0
    THEN average = sum / total.valid;
    ELSE average = -999;
ENDIF

END average
```

2.2 Given Diagram (Use as the CFG reference)

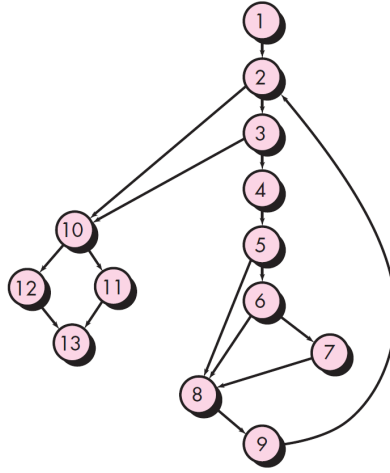


Figure 1: Given pseudocode with labeled points (1–13) used as CFG nodes.

3 Step 1: Identify Graph Counts (N, E, P, R, P_d)

From the numbered diagram, we treat the labeled circles/points as CFG nodes:

$$N = 13, \quad P = 1$$

(one connected component).

3.1 Edges E

Edge listing (consistent with the flow shown in the diagram):

- Main flow: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 9$
- Loop back-edge: $9 \rightarrow 2$
- Edges to the final decision: $10 \rightarrow 11$, $10 \rightarrow 12$, $11 \rightarrow 13$, $12 \rightarrow 13$
- Links from early checks to the final block: $2 \rightarrow 10$, $3 \rightarrow 10$
- “short-circuit/skip” style edges in the validation part: $5 \rightarrow 8$, $6 \rightarrow 8$

So the total edges are:

$$E = 17.$$

3.2 Predicate Nodes P_d

A **predicate node** is a node with a decision (i.e., more than one outgoing edge).

From the diagram structure, decision points occur at:

- Node 2 (loop condition / can exit to 10 or continue)
- Node 3 (compound loop check / can exit to 10 or continue)
- Node 5 (validation check can go to 6 or skip to 8)
- Node 6 (compound validation check can go to 7 or skip to 8)
- Node 10 (final decision splits to 11 or 12)

Hence,

$$P_d = 5.$$

3.3 Regions R

Counting enclosed areas in the CFG (including the outside region) gives:

$$R = 6.$$

4 Step 2: Calculate Cyclomatic Complexity $V(G)$ (All Methods)

Method 1 (Regions):

$$V(G) = R = 6$$

Method 2 (Edges–Nodes):

$$V(G) = E - N + 2P = 17 - 13 + 2(1) = 6$$

Method 3 (Predicate Nodes):

$$V(G) = P_d + 1 = 5 + 1 = 6$$

$$V(G) = 6$$

So, **6 independent (basis) paths** are required for Basis Path Testing.

5 Step 3: Identify a Basis Set of Independent Paths

A **basis path** is an **independent path** that introduces at least one new edge not included in any other selected path.

One valid basis set (6 independent paths) is:

1. $1 \rightarrow 2 \rightarrow 10 \rightarrow 11 \rightarrow 13$

2. $1 \rightarrow 2 \rightarrow 10 \rightarrow 12 \rightarrow 13$
3. $1 \rightarrow 2 \rightarrow 3 \rightarrow 10 \rightarrow 11 \rightarrow 13$
4. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 8 \rightarrow 9 \rightarrow 2 \rightarrow 10 \rightarrow 11 \rightarrow 13$
5. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 8 \rightarrow 9 \rightarrow 2 \rightarrow 10 \rightarrow 11 \rightarrow 13$
6. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 2 \rightarrow 10 \rightarrow 11 \rightarrow 13$

These 6 paths collectively cover the loop exit/continue logic, the validation skip/accept logic, and both outcomes of the final decision at node 10.

6 Step 4: Minimum Test Cases

Minimum number of test cases for full basis path coverage:

$$\text{Test Cases} = V(G) = 6$$